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**Optimising Fast Radio Burst Search Pipelines: A
Comparative Study of Digital Image Processing
Filters for Structural Morphology Preservation**

SUBMITTED BY

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1.1 Introduction

Fast Radio Bursts

Fast Radio Bursts (FRBs) are highly energetic, millisecond-duration radio transients originating from extragalactic distances. Since their first discovery in 2007, FRBs have become one of the most intriguing phenomena in modern radio astronomy due to their unknown origins and immense energy output. Although these bursts last only a few milliseconds, they can release energy comparable to that emitted by the Sun over several days, making them valuable probes for studying the distant Universe.

Scientific Significance

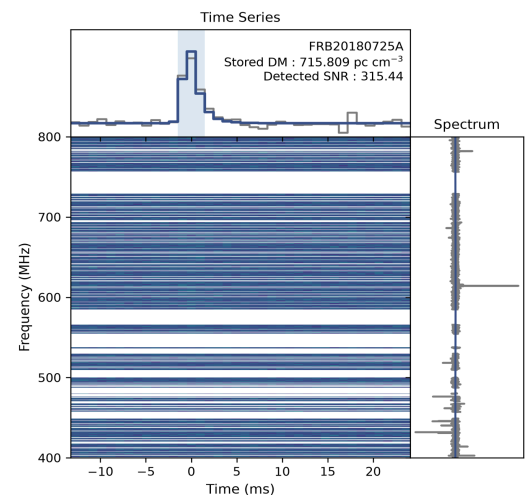
As radio waves from an FRB travel through the interstellar and intergalactic medium, they undergo frequency-dependent dispersion caused by free electrons along their propagation path. Consequently, lower-frequency components arrive later than higher-frequency components, producing a characteristic dispersed signal. This dispersion provides important information about the distribution of ionised matter between the source and the observer, enabling studies of the large-scale structure of the Universe, baryonic matter distribution, cosmic magnetic fields, and the environments surrounding compact astrophysical objects.

FRB Observations

Modern radio observatories, such as the **Canadian Hydrogen Intensity Mapping Experiment (CHIME)**, have revolutionised FRB discovery by continuously monitoring large regions of the sky with high temporal and spectral resolution. The recorded observations are stored as **dynamic spectra**, commonly known as **waterfall plots**, in which signal intensity is represented as a function of time and observing frequency. Unlike conventional one-dimensional time-series data, these two-dimensional representations preserve the complete burst morphology, including temporal evolution, frequency-dependent behaviour, scattering tails, and other fine structural characteristics essential for scientific analysis.

Need for Effective Preprocessing

Despite their scientific importance, raw FRB observations are often contaminated by instrumental noise, receiver gain fluctuations, baseline drift, and terrestrial **Radio Frequency Interference (RFI)**. These artefacts reduce burst visibility and complicate reliable signal detection and characterisation. Therefore, effective preprocessing is essential to suppress unwanted noise while preserving the intrinsic morphology of the burst. Achieving this balance is critical for accurate burst detection, feature extraction, and subsequent astrophysical analysis.



1.2 Motivation and Research Gap

The increasing number of Fast Radio Burst (FRB) detections has shifted the focus of modern radio astronomy from event discovery to accurate burst characterisation. Large-scale surveys such as CHIME continuously generate high-resolution time–frequency observations, creating a need for automated processing frameworks that are both computationally efficient and scientifically reliable.

Conventional 1D Processing

Current FRB search pipelines are optimised for real-time detection by collapsing the two-dimensional dynamic spectrum into a one-dimensional time series. Although computationally efficient, this transformation discards valuable information such as frequency-dependent burst evolution, scattering tails, burst asymmetry, and fine morphological structures.

Information Loss

The conversion from a 2D waterfall image to a 1D signal creates a trade-off between processing speed and information preservation. Weak, scattered, or morphologically complex bursts become more difficult to detect and characterize, often resulting in reduced sensitivity and inaccurate peak localization.

Noise and Instrumental Artifacts

Raw FRB observations are affected by:

- Baseline drift and bandpass variations
- Receiver gain fluctuations
- Thermal noise
- Radio Frequency Interference (RFI)

These artefacts obscure astrophysical signals and reduce the reliability of automated burst detection.

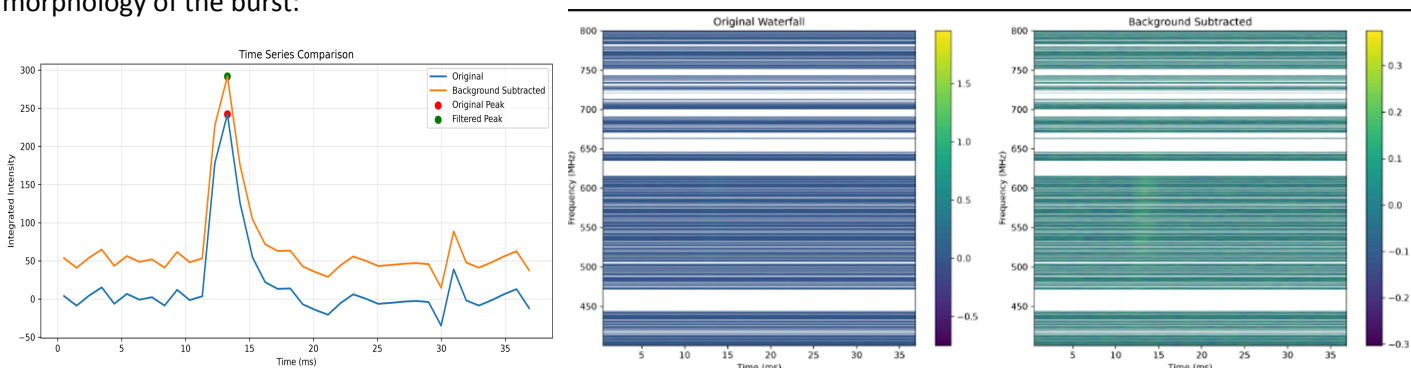
Limitations of Conventional Denoising

Traditional preprocessing methods primarily focus on suppressing noise but often introduce:

- Pulse broadening
- Peak attenuation
- Morphological distortion
- Loss of high-frequency structural information

Such distortions directly impact burst detection, segmentation, feature extraction, and subsequent scientific analysis.

Example: simply using the background subtraction method increased the SNR but without preserving the morphology of the burst:



Need for a Morphology-Aware Framework

These limitations motivate the development of a preprocessing framework that operates directly on the two-dimensional dynamic spectrum rather than the collapsed one-dimensional signal. By applying Digital Image Processing techniques to the original waterfall images, it becomes possible to suppress instrumental artefacts while preserving the intrinsic burst morphology. This provides a stronger foundation for reliable burst detection, quantitative feature extraction, and the development of robust next-generation FRB search pipelines.

1.3 Objectives of the Study

The primary objective of this work is to develop a reproducible offline image-processing framework for Fast Radio Burst (FRB) dynamic spectra that enhances burst detectability while preserving the intrinsic structural morphology of the signal. By combining Digital Image Processing techniques with quantitative performance evaluation, the study aims to investigate how different preprocessing methods influence burst characterisation and to identify an optimised workflow suitable for automated FRB analysis.

The specific objectives of this study are:

- To develop a modular and reproducible preprocessing pipeline for FRB dynamic spectra obtained from the CHIME/FRB Open Data Catalogue.
- To investigate the effectiveness of multiple Digital Image Processing techniques for suppressing instrumental noise and Radio Frequency Interference (RFI) while preserving astrophysically significant burst features.
- To perform a systematic comparative analysis of commonly used denoising filters, including Median, Gaussian, Wiener, Bilateral, Wavelet, Histogram Equalisation, and CLAHE, using quantitative image quality and signal preservation metrics.
- To evaluate the influence of different preprocessing methods on burst morphology, signal-to-noise ratio, structural similarity, and feature preservation across a large sample of FRB events.
- To implement automated burst detection, segmentation, and feature extraction techniques for objective characterisation of FRB dynamic spectra.
- To benchmark the performance of the proposed preprocessing framework using statistical evaluation methods and identify the most suitable denoising strategy for downstream burst analysis.
- To develop an optimised morphology-aware processing workflow that provides a reliable foundation for future applications such as automated burst classification, machine learning-based detection, and improved FRB search pipelines.

1.4 Dataset

The study utilises publicly available observations from the CHIME/FRB Open Data Catalogue, which provides calibrated observations of confirmed Fast Radio Burst (FRB) events along with standardised data products and observational metadata. [CHIME/FRB Open Data Catalogue](#)

Dataset Summary

- Source: CHIME/FRB Open Data Catalogue
- Total Events: 535 independent and authenticated FRB events
- Data Format: Hierarchical Data Format (HDF5, .h5)
- Primary Data Product: Dedispersed dynamic spectra (waterfall images)
- Data Representation: Two-dimensional time-frequency intensity maps preserving burst morphology and spectral evolution

Dataset Characteristics

The selected dataset includes realistic observational conditions containing:

- Thermal receiver noise
- Baseline and bandpass variations
- Residual Radio Frequency Interference (RFI)
- Localised intensity fluctuations

These characteristics provide a realistic benchmark for evaluating preprocessing and denoising techniques.

Evaluation Framework

- All FRB events were processed using an identical preprocessing workflow.
- Each denoising method was applied independently using the same processing parameters.
- Feature extraction and performance evaluation were performed using a common analysis pipeline.
- A standardised evaluation framework ensured fair and unbiased comparison across all methods while minimising implementation bias.

Dataset Significance

The dataset encompasses a diverse range of FRB properties, including variations in:

- Signal-to-noise ratio (SNR)
- Burst brightness
- Temporal width
- Spectral occupancy
- Scattering characteristics
- Burst morphology

This diversity makes the dataset well suited for benchmarking Digital Image Processing techniques and developing a robust, morphology-preserving preprocessing framework for large-scale FRB analysis.

1.5 Problem Diagnosis

Although the native CHIME offline pipeline is highly effective for large-scale real-time FRB detection, its design prioritises computational efficiency over preservation of burst morphology. Several limitations motivate the development of an improved preprocessing framework.

Skipping Baseline Detrending

- Residual baseline drift and bandpass variations remain in the dynamic spectrum.
- These variations reduce burst contrast and affect reliable signal extraction.

Rigid Threshold-Based RFI Detection

- Fixed thresholding techniques struggle to identify localised and dynamically varying Radio Frequency Interference (RFI).
- Weak bursts embedded within interference may be partially suppressed or completely missed.

Symmetric Boxcar Detection

- Conventional boxcar filters assume symmetric burst profiles.
- Many FRBs exhibit asymmetric structures and scattering tails that are not well represented by boxcar templates.

Arrival-Time Bias

- Symmetric filtering can broaden the pulse and shift the estimated burst peak.
- This introduces timing uncertainties that affect subsequent parameter estimation.

Need for an Improved Framework

These limitations motivate the development of a morphology-aware preprocessing framework that combines adaptive preprocessing, image-based denoising, and optimised burst detection while preserving the intrinsic structure of FRB signals.

2. Pipeline Architecture

2.1 Common Front-End Preprocessing

Both Pipeline A and Pipeline B are built on a shared front-end that replaces the Native portal's uncorrected processing with three deliberate corrective stages, applied before the pipelines diverge.

Local Window Detrending. Slow bandpass and baseline drift is removed on a cell-by-cell basis across the dynamic spectrum (waterfall), while fast, millisecond-scale burst structure is preserved. This directly addresses the Native pipeline's “*skip detrending*” step, which rides blindly on an uncleaned, drifting raw noise floor.

Adaptive MAD RFI Masking. A per-file, scaled threshold is estimated from the median absolute deviation (MAD) of the data and used to flag and mask radio-frequency interference. For a channel value x , the robust noise scale is

$$\hat{\sigma}_{\text{MAD}} = 1.4826 \times \text{median}(| x_i - \text{median}(x) |)$$

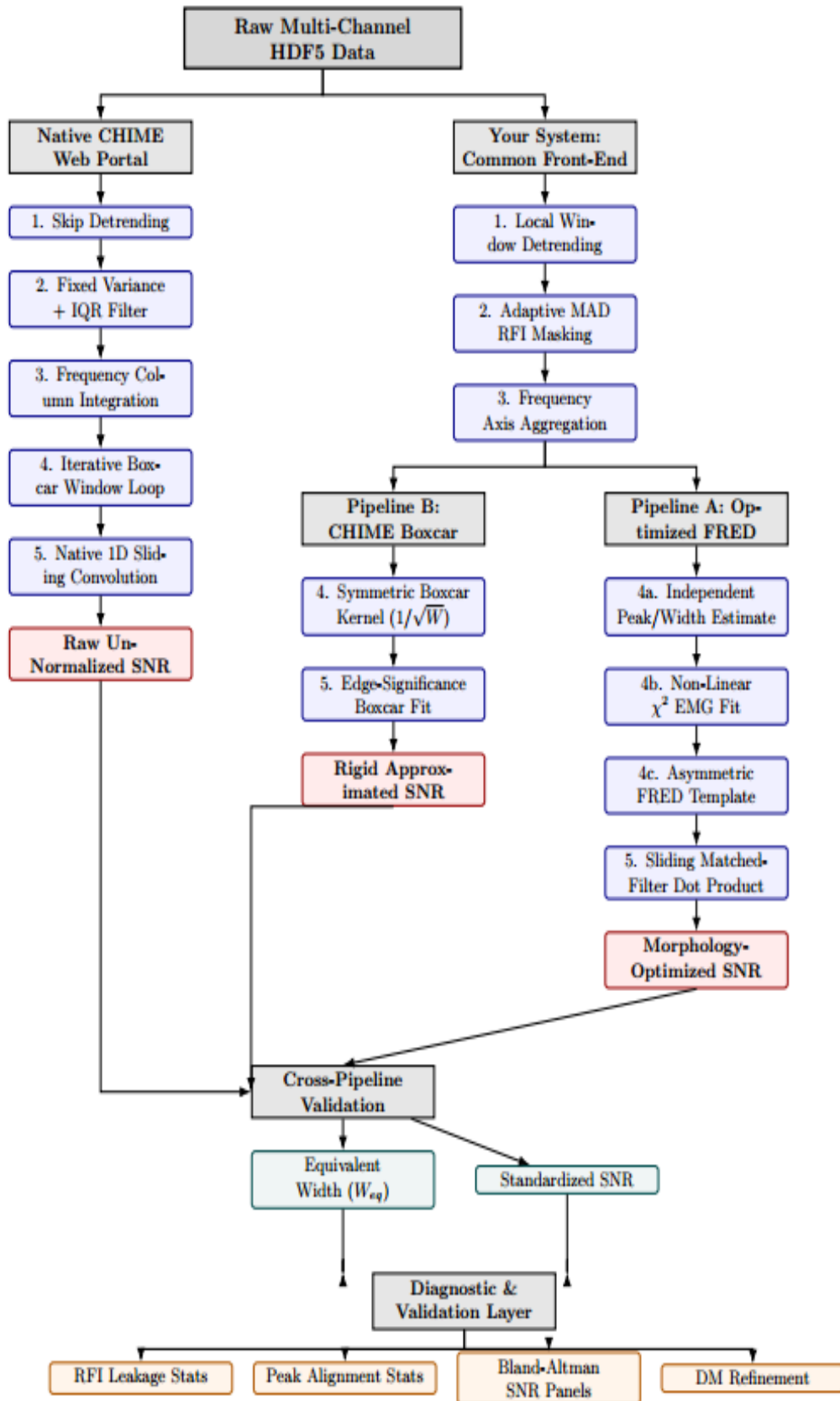
with the masking threshold set as a multiple of $\hat{\sigma}_{\text{MAD}}$ re-estimated separately for every file. This replaces Native's fixed-threshold filter,

$$\sigma^2 > 3\langle \sigma^2 \rangle$$

which uses a single, non-adaptive arithmetic-mean-based cut and consequently exhibits masking blindness on files whose noise properties deviate from the assumed average.

Frequency-Axis Aggregation. The cleaned, detrended matrix is collapsed from 16,384 fine-frequency channels into a single one-dimensional time series by summing across the frequency axis — the same conceptual step as Native's column integration, but operating on cleaned rather than raw data.

CHIME Offline Pipeline Architecture (Flowchart)



2.2 Pipeline A Optimised Asymmetric FRED

Pipeline A extends the common front-end with a morphology-aware fitting chain designed to match the physical shape of a fast radio burst rather than approximate it with a generic template.

(a) Independent Peak/Width Estimate. An initial peak location and width are estimated directly from Pipeline A's own cleaned time series, not inherited from Pipeline B's boxcar output — this ensures that any bias in B's estimate cannot leak into A's fit.

(b) Non-Linear χ^2 Model Fit. The burst is modelled as an Exponentially Modified Gaussian (EMG), the convolution of a Gaussian intrinsic pulse with an exponential scattering tail:

$$\text{EMG}(t) = \text{Gauss}(\sigma) * \text{Exp}(\tau)$$

where σ is the intrinsic Gaussian width and τ is the exponential scattering timescale. Explicitly,

$$\text{EMG}(t) = A \cdot \exp(\sigma^2 / 2\tau^2 - (t - t_0) / \tau) \cdot \text{erfc}([(\sigma/\tau) - (t-t_0)/\sigma] / \sqrt{2})$$

The parameters σ and τ are fit nonlinearly via χ^2 minimisation, thereby decoupling the intrinsic pulse width from scattering-induced broadening. A dual-EMG model (sum of two EMG components) is used when the burst shows two resolvable sub-pulses.

(c) Asymmetric FRED Template. The fitted EMG parameters are used to construct an energy-normalised, Fast-Rise-Exponential-Decay (FRED) template moulded to the burst's actual asymmetric morphology, rather than assuming a symmetric pulse shape.

(d) Sliding Matched-Filter Dot Product. The normalised template is cross-correlated against the time series via a sliding dot product, and the peak of this cross-correlation is taken as the burst arrival time. This fixes the centre-of-mass drag introduced by simple convolution-based peak-finding (as used by the Native pipeline via `scipy.signal.convolve`). The resulting Morphology-Optimized SNR is

$$\text{SNR}_A = \max(t_s(t) \cdot \text{EMG}_{\text{norm}}(t))$$

where $t_s(t)$ is the cleaned, aggregated time series and EMG_{norm} is the energy-normalised template.

2.3 Pipeline B — CHIME Boxcar Baseline

Pipeline B represents an improved but conventional approach, retained as an intermediate baseline between the Native portal and Pipeline A.

Symmetric Boxcar Kernel. A flat top-hat kernel of trial width W is convolved with the time series, with amplitude scaled as

$$\text{SNR}(W) \propto 1 / \sqrt{W}$$

across all trial widths, following standard matched-filtering-for-top-hat practice.

Edge-Significance Boxcar Fit. Candidate widths are rejected when both edges of the boxcar sit near the noise floor — a partial correction for “tail sponging,” the failure mode where an overly wide boxcar accumulates scattering-tail noise and artificially inflates its own significance. This is an improvement over

Native's uncorrected iterative boxcar loop, but remains a rigid, symmetric approximation of what is frequently an asymmetric pulse profile.

2.4 Cross-Pipeline Validation Layer

To compare pipelines that operate in different internal units, two shared, pipeline-agnostic metrics are computed downstream of all three:

Equivalent Width.

$$W_{eq} = \text{Fluence} / \text{Peak Amplitude}$$

computed from each pipeline's own (peak, baseline) pair, making the comparison genuinely like-for-like rather than an artefact of differing calibration.

Standardised SNR.

$$\text{SNR}_{std} = \text{raw amplitude} / \sigma_{\text{off-pulse}}$$

which expresses every pipeline's amplitude in units of its own off-pulse noise σ , putting Native, Pipeline B, and Pipeline A on the same sigma-normalised scale despite their different preprocessing. A read-only diagnostic layer is computed alongside these: RFI leakage statistics (skewness/excess kurtosis of off-pulse noise, per masking branch), peak-alignment statistics (offset of each pipeline's fitted peak vs. a raw reference), Bland–Altman panels (pairwise mean-vs-difference plots for both raw and standardised SNR).

Code Availability

All source code, processing scripts, and optimisation pipelines developed in this work are publicly available on GitHub at: https://github.com/nischayaj/CHIME_Pipeline_optimization

3. Burst Detection and Noise Reduction Methods

Following calibration and standard CHIME/FRB preprocessing, the dynamic spectra of 535 independent Fast Radio Burst (FRB) events were analyzed using a two-dimensional image-domain processing framework designed to enhance burst detectability while preserving intrinsic time–frequency morphology. Although calibration, baseline correction, radio frequency interference (RFI) mitigation, and frequency reduction improve data quality, residual fluctuations caused by receiver noise, calibration uncertainties, instrumental variations, and weak interference components may remain. These residual structures can affect signal recovery and introduce uncertainties in derived properties such as signal-to-noise ratio (SNR), burst width, spectral extent, and emission morphology [1, 2].

Unlike conventional FRB search pipelines that primarily operate on frequency-integrated one-dimensional time series, this work treats calibrated waterfall data as two-dimensional time–frequency images. Dynamic spectra preserve both temporal evolution and frequency-dependent intensity variations, allowing image-processing methods to exploit local correlations across both dimensions while maintaining physically meaningful burst structures [3].

All FRB dynamic spectra underwent identical preparation before application of the evaluated preprocessing methods. The frequency dimension was reduced using a binning factor of 16 to improve computational efficiency while retaining dominant burst features. Frequency downsampling was performed using NaN-aware averaging, and remaining undefined pixels were replaced using observation-specific median intensity values. The same prepared inputs were provided to all algorithms to ensure that performance differences originated from the applied methods rather than variations in initial preprocessing.

3.1 Denoising and Contrast Enhancement Methods

Six denoising techniques representing different filtering strategies were evaluated: Median, Gaussian, Wiener, Boxcar, Bilateral, and Wavelet filtering. Median filtering was implemented as a nonlinear order-statistic filter with a kernel size of (3,1). Each pixel intensity was replaced by the median value within the local neighborhood, reducing impulsive fluctuations while preserving sharp burst transitions.

Gaussian filtering was applied using Gaussian convolution with $\sigma = 1$:

$$G(x, y) = (1 / 2\pi\sigma^2) e^{-(x^2 + y^2) / (2\sigma^2)}$$

where σ controls the smoothing strength.

Wiener filtering was implemented using adaptive local noise estimation with a window size of (3,3), estimating local image statistics to suppress noise while preserving spatial variations.

Boxcar filtering was applied using uniform averaging with a window size of (3,3). While effective for reducing random fluctuations, this approach can smooth localized burst structures.

Bilateral filtering was evaluated as an edge-preserving smoothing technique using both spatial and intensity similarity constraints. The implementation used $\sigma_{\text{color}} = 0.05$ and $\sigma_{\text{spatial}} = 3$.

Wavelet denoising was performed using multiresolution wavelet decomposition with BayesShrink threshold estimation. The implementation used soft thresholding with rescaled noise estimation.

Wavelet Denoising Parameters:

Thresholding method: BayesShrink

Thresholding mode: Soft thresholding

Sigma rescaling: Enabled (True)

The wavelet basis and decomposition level were kept at the default values of the scikit-image implementation. This approach suppresses noise-dominated wavelet coefficients while retaining coefficients associated with burst emission structures [4, 5].

In addition to the denoising methods, Histogram Equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE) were evaluated separately as contrast enhancement techniques. Histogram Equalization redistributes global intensity values, while CLAHE performs local contrast enhancement with a clip limit of 0.03. Since these methods modify intensity distributions rather than directly suppress stochastic noise, they were analyzed independently from the primary denoising comparison.

3.2 Quantitative Evaluation Framework

The performance of each preprocessing method was evaluated using metrics describing signal enhancement, noise behavior, structural preservation, burst morphology, image quality, and computational efficiency. Peak SNR was calculated from the burst peak intensity relative to the surrounding background noise:

$$\text{SNR}_{\text{peak}} = (I_{\text{peak}} - \mu_{\text{noise}}) / \sigma_{\text{noise}}$$

where I_{peak} represents the intensity at the burst peak, μ_{noise} is the mean background noise estimated outside the burst region, and σ_{noise} is the corresponding noise standard deviation.

Integrated SNR was calculated over the burst region:

$$\text{SNR}_{\text{int}} = (\sum S - \mu_{\text{noise}} \times N) / (\sigma_{\text{noise}} \times \sqrt{N})$$

where ΣS represents the summed burst intensity, N is the number of samples within the burst window, and μ_{noise} and σ_{noise} are estimated from off-burst regions.

Noise reduction was quantified using background RMS variation:

$$\text{Noise Reduction (\%)} = ((\sigma_{\text{baseline}} - \sigma_{\text{processed}}) / \sigma_{\text{baseline}}) \times 100$$

where:

σ_{baseline} = standard deviation of noise before processing

$\sigma_{\text{processed}}$ = standard deviation of noise after applying the denoising method

Peak preservation was calculated as:

$$\text{Peak Preservation (\%)} = (I_{\text{processed_peak}} / I_{\text{baseline_peak}}) \times 100$$

where:

$I_{\text{processed_peak}}$ = peak intensity after applying the processing method

$I_{\text{baseline_peak}}$ = original peak intensity before processing

Burst morphology preservation was assessed using FWHM, burst duration, spectral centroid, occupied bandwidth, spatial extent, equivalent diameter, and convex area measurements.

Image quality was evaluated using Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \times \log_{10}(\text{MAX}_i^2 / \text{MSE})$$

where:

MAX_i = maximum possible pixel intensity value of the image

MSE = mean squared error between the original and processed image

and Structural Similarity Index (SSIM), which measures luminance, contrast, and structural consistency between processed and reference images [6].

3.3 Statistical Validation

Since every FRB event was processed using all denoising approaches, the resulting measurements formed a repeated-measures dataset. As several metrics deviated from normality, non-parametric statistical analysis was performed. The Friedman test was used to determine whether significant differences existed among denoising approaches. Significant results were followed by Wilcoxon signed-rank pairwise comparisons with Bonferroni correction to identify differences between individual methods [7].

This framework combines two-dimensional image-domain processing, multi-metric evaluation, and statistical validation to identify preprocessing strategies that optimise FRB signal recovery and preserve intrinsic time–frequency morphology.

4. Results

4.1 Pipeline Metrics

Standardising all three pipelines' SNR to units of off-pulse σ removes each pipeline's own noise-floor calibration as a confound, so amplitude gains can be attributed to the detection method rather than to differing normalisation.

Pipeline A achieves a **mean standardized-SNR ratio of 1.39 $\times$ over Native** (versus 1.12 $\times$ for Pipeline B), indicating that the FRED/matched-filter approach recovers substantially more signal on average than either the naive portal method or the boxcar baseline.

The **median ratios — 1.18 $\times$ for A, 0.99 $\times$ for B —** are the more diagnostic figures for assessing systematic bias. A median near 1.0 for Boxcar-vs-Native shows Pipeline B is essentially bias-neutral relative to Native for a typical burst. Pipeline A's median of 1.18 $\times$ confirms its improvement is real and not solely an artefact of a small number of very bright outliers — though, as discussed in Section 4.3, the gap between its mean and median ratio is itself evidence that the gain is unevenly distributed across the catalogue.

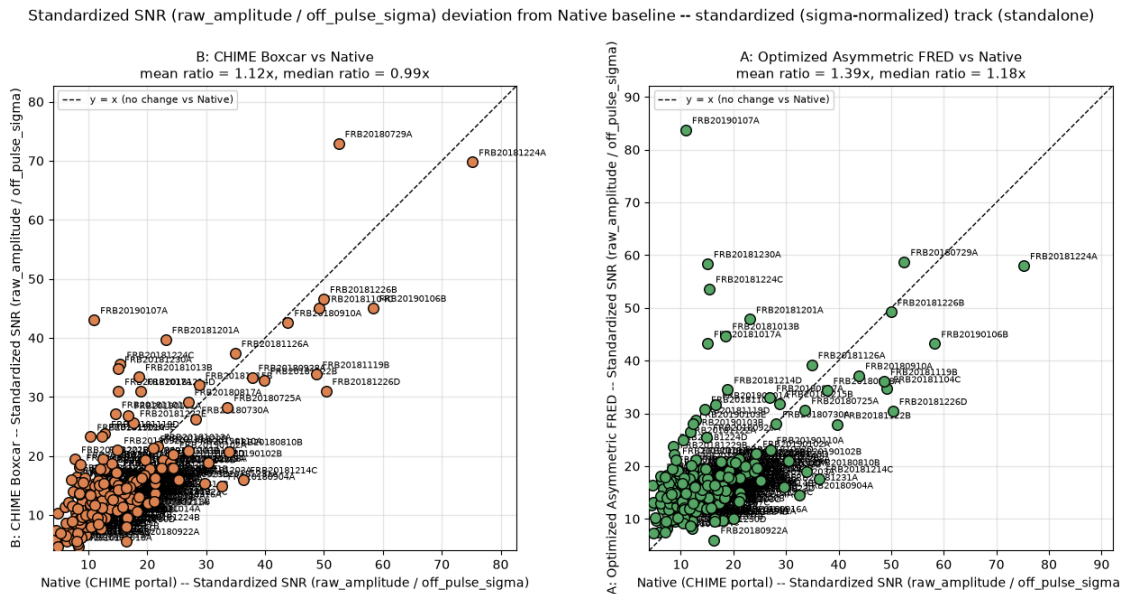


Figure 1. Standardised SNR of Pipeline B (left) and Pipeline A (right) vs. Native, per burst. Points above the $y = x$ line indicate improvement over Native; the mean/median ratios reported above are computed from this comparison.

4.2 Cross-Validation Synthesis — the KS Test

The SNR comparison above was run on a 200-event subset of the full 534-event catalogue rather than on every burst. This introduces a specific risk: if the subset happens to be skewed toward brighter, closer, or less-scattered bursts, the reported SNR gains would reflect a property of the sample rather than of the pipeline.

The two-sample Kolmogorov–Smirnov (KS) test was run to rule this out, comparing the dispersion measure (DM) distribution of the full catalogue against the 200-event subset. DM is used as the test variable because it is a strong proxy for burst distance, scattering environment, and expected signal degradation if the subset's DM distribution differed significantly from the full catalogue's, the measured SNR gains could not be trusted as representative of the catalogue as a whole.

The KS statistic is defined as the maximum vertical distance between the two empirical cumulative distribution functions:

$$D = \sup_x | F_1(x) - F_2(x) |$$

where F_1 and F_2 are the empirical CDFs of the full catalogue and the subset, respectively. As shown in Figure 2, the two curves are visually near-identical across the entire 0–3000 pc cm⁻³ range, and the resulting KS statistic is small enough that the null hypothesis that both samples are drawn from the same underlying DM distribution cannot be rejected.

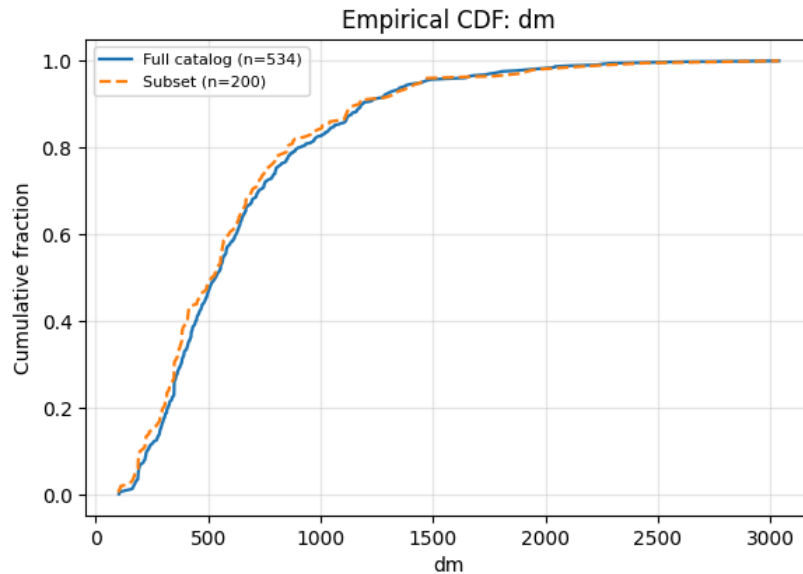


Figure 2. Empirical CDF of DM for the full catalogue ($n = 534$, solid) vs. the 200-event subset (dashed) used for SNR comparison. Close agreement supports the KS test conclusion that the subset is not a biased draw.

This means the mean/median SNR ratios reported in Section 4.1 can reasonably be generalised to the full catalogue, rather than being an artefact of which 200 events happened to be selected for the standardized-SNR comparison.

4.3 Limitations and Negative Results

The following weaknesses are reported directly, as they materially qualify the headline SNR improvement above.

- **Pipeline A's gains are outlier-driven, not uniform.** The gap between the mean ratio (1.39×) and median ratio (1.18×) in Figure 1 is itself evidence of this: a handful of bright bursts (e.g. FRB20190107A, FRB20181230A, FRB20181224C) reach 50–80× standardised SNR and pull the mean upward, while many points sit close to, or even below, the $y = x$ line. For a typical faint burst, the improvement over Native is real but modest, not the 1.39× the mean alone would suggest.
- **Pipeline A shows worse peak-timing outliers than Pipeline B.** In the peak-alignment comparison (Figure 4, centre panel), Pipeline B's offsets cluster tightly near 0 ms with few outliers, whereas Pipeline A shows several events offset by 30–40 ms from the raw reference, including one at −30 ms. The FRED template occasionally locks onto a scattering tail or a secondary sub-pulse instead of the true peak a failure mode the simpler, more rigid boxcar rarely exhibits. This is a genuine trade-off: morphology-awareness buys SNR at the cost of occasional timing misfires. *An example of the outliers is shown in Figure 5 along with the typical set.*
- **Adaptive MAD masking improves the average case but not the worst case.** Figure 3 (left panel) and Figure 4 both show the bulk of points tightening up under the adaptive threshold relative to Native's fixed threshold, but the single most extreme RFI outlier in the catalogue (tagged “a”) is essentially unchanged between the two masking branches. The adaptive approach reduces typical-file RFI leakage but does not protect against pathological single-file contamination.
- **Cross-pipeline agreement is heteroscedastic (unequal scatter).** The Bland–Altman panels in Figure 4 (top-right and bottom row) show the spread of standardized-SNR differences widening as mean SNR increases, rather than remaining constant. Pipeline disagreement grows for brighter bursts the opposite of what would be expected from a uniformly reliable cross-pipeline calibration.
- **Pipeline B shows essentially no net benefit over Native for typical bursts.** Its median ratio of 0.99× (Figure 1, left panel) means that for roughly half the catalogue, the “improved” boxcar method with edge-significance fitting performs no better than the naive tutorial baseline. The entire measured gain for Pipeline B comes from a small number of high-SNR events, not from a systematic improvement in typical-case sensitivity.

RFI Leakage: Skewness vs. Excess Kurtosis -- decoupled per RFI-masking branch

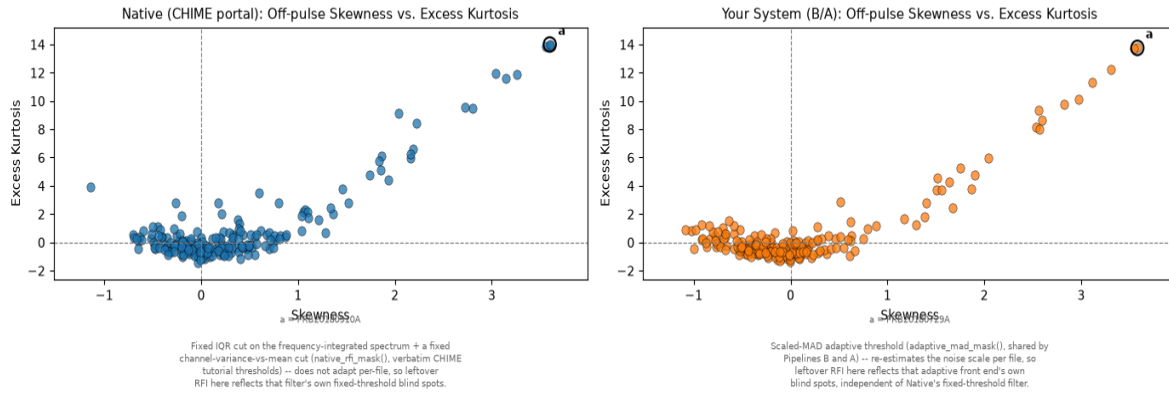


Figure 3. Off-pulse skewness vs. excess kurtosis, decoupled by RFI-masking branch (Native fixed-threshold, left; adaptive-MAD front end shared by B/A, right). The shared outlier “a” appears in both panels, showing it is not resolved by the adaptive threshold.

Diagnostic Summary Dashboard -- Standardized SNR (additive, read-only view of existing metrics)

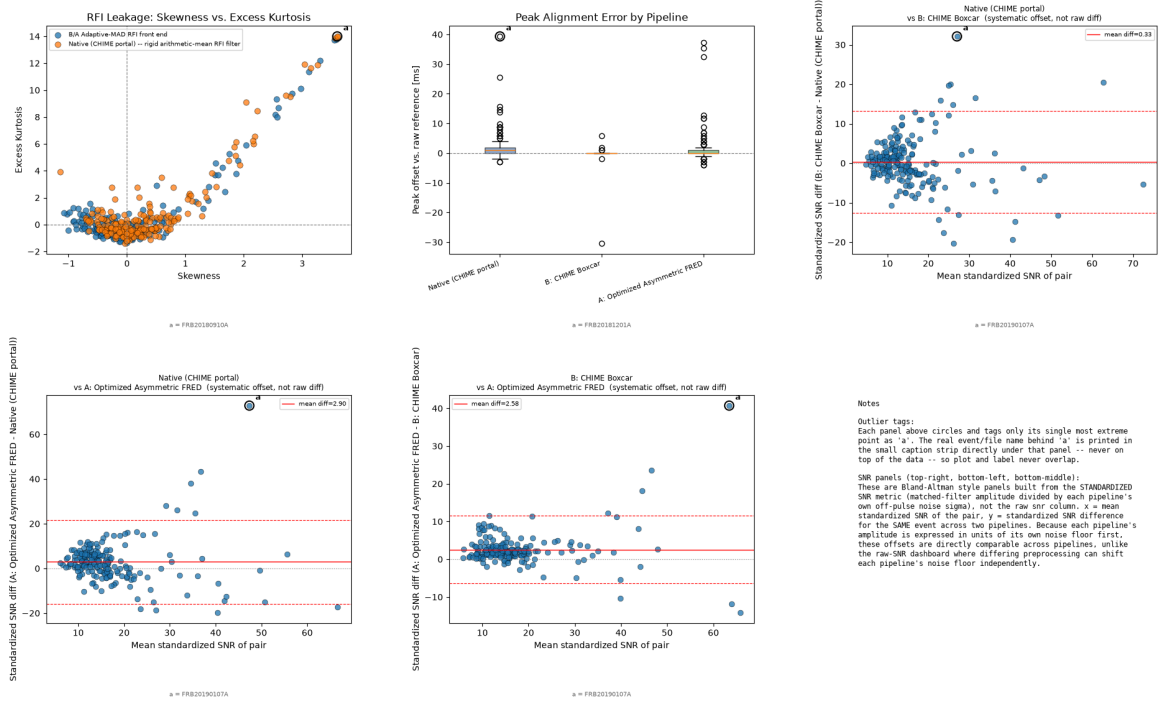
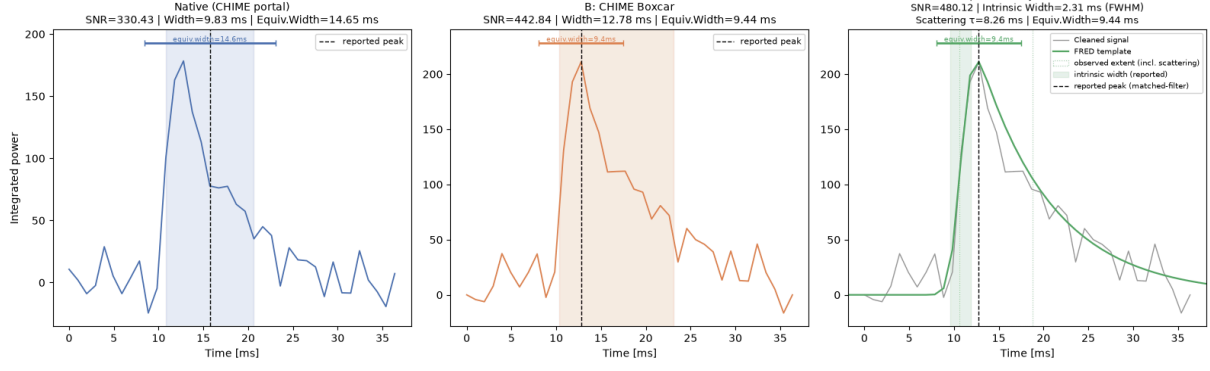
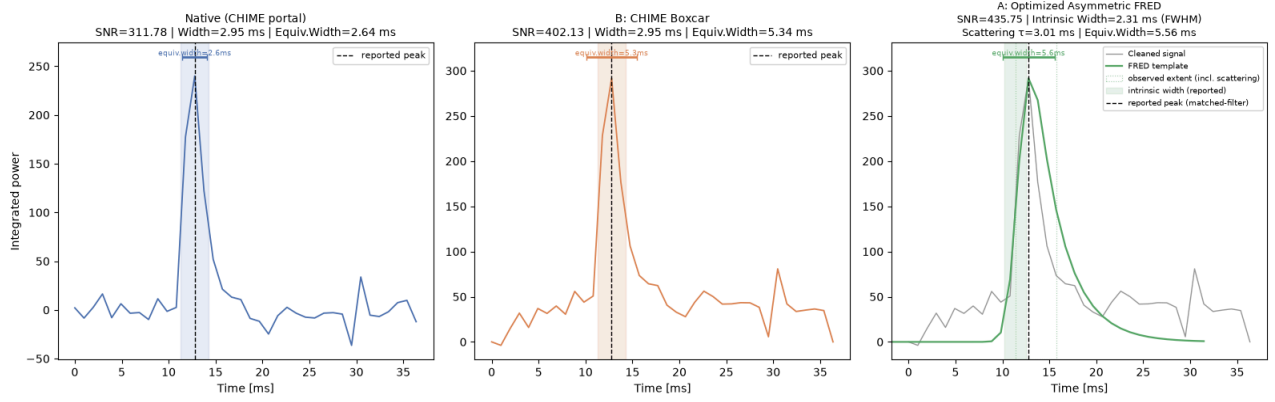


Figure 4. Diagnostic summary dashboard: RFI leakage (top-left), peak-alignment error by pipeline (top-centre), and Bland–Altman standardized-SNR panels for all three pairwise comparisons. Widening spread at higher mean SNR (right-hand panels and bottom row) indicates heteroscedastic cross-pipeline agreement.

Multi-Pipeline Signal Profile -- FRB20190701D



Multi-Pipeline Signal Profile -- FRB20180725A



Multi-Pipeline Signal Profile -- FRB20180814A

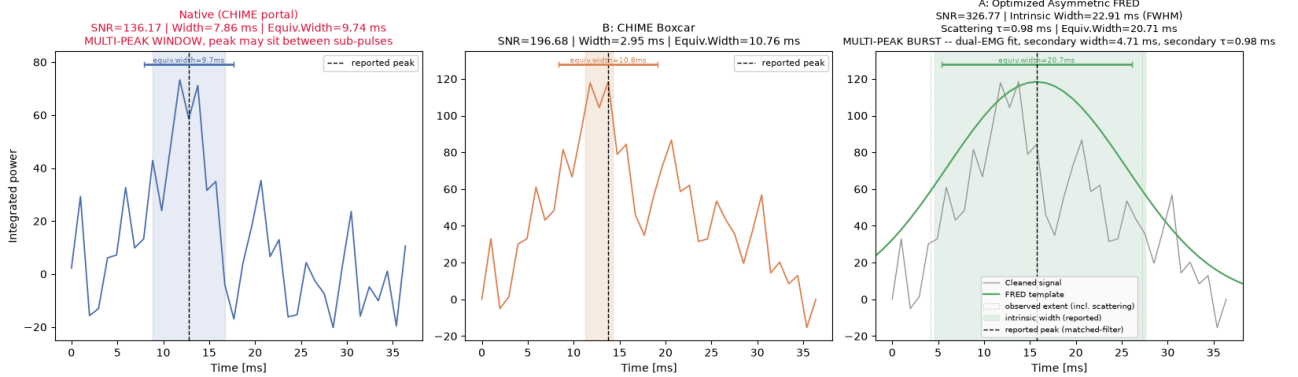


Figure 5. Shows the set of examples that were derived from the pipelines.(FRB20190701D, FRB20180814A are examples of outliers; here we see that pipeline B is performing relatively well. The dual peaks arose due to multipath scattering/properagation while the radiation was travelling through interstellar space)

4.4 Comparative Performance of Preprocessing Methods

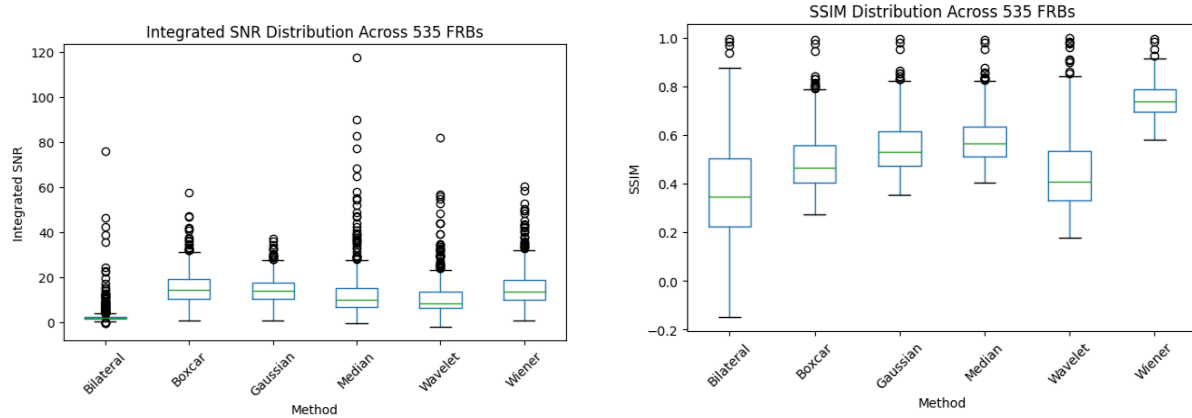
The performance of nine preprocessing approaches was evaluated across 535 independent FRB dynamic spectra using the multi-metric framework described in Section 3. The evaluated methods included six denoising techniques (Median, Gaussian, Wiener, Boxcar, Bilateral, and Wavelet), two contrast enhancement techniques (Histogram Equalization and CLAHE), and the original baseline. The comparison considered signal enhancement, noise suppression, structural preservation, burst morphology retention, and computational performance.

Adaptive denoising techniques outperformed conventional smoothing approaches. Wavelet filtering achieved the highest signal enhancement, with Peak SNR = 29.37, Integrated SNR = 52.65, and noise reduction = 60.49%, demonstrating effective separation of stochastic background fluctuations from localized burst emission. Wiener filtering also showed strong performance, achieving Peak SNR = 24.22 and Integrated SNR = 30.62. Boxcar filtering improved Integrated SNR (32.69) but introduced greater smoothing of localized burst structures due to spatial averaging. Gaussian and Median filtering provided moderate improvements.

Histogram Equalisation and CLAHE improved intensity contrast but did not directly suppress noise, as these methods modify intensity distributions rather than remove background fluctuations. Therefore, they are more suitable as supplementary visualisation or post-processing techniques rather than standalone denoising methods.

Computational analysis further revealed differences in processing efficiency among preprocessing techniques, highlighting the trade-off between enhancement capability, structural preservation, and computational cost.

Overall, adaptive denoising approaches, particularly Wavelet and Wiener filtering, provided the most effective enhancement for FRB dynamic spectra containing localized signals embedded within complex two-dimensional noise backgrounds.



Fig(left) - Boxplot of Integrated SNR for all 535 FRBs with all the filters. (Right)- Boxplot of SSIM for all 535 FRBs with all the filters.

4.5 Structural and Morphological Preservation

The impact of preprocessing on FRB structural characteristics was evaluated using PSNR, SSIM, and extracted burst morphology parameters, including FWHM, burst duration, spectral centroid, occupied bandwidth, spatial extent, equivalent diameter, and convex area.

Wiener filtering provided the strongest balance between enhancement and structural preservation, achieving PSNR = 25.74 dB and SSIM = 0.7325. Median and Gaussian filtering showed moderate preservation with PSNR = 22.69 dB, SSIM = 0.5279 and PSNR = 22.85 dB, SSIM = 0.4804, respectively. Boxcar smoothing produced lower structural similarity (PSNR = 22.19 dB, SSIM = 0.4139), indicating loss of localized burst information due to uniform averaging.

Although Wavelet filtering produced the highest signal recovery, its lower structural similarity (PSNR = 21.90 dB, SSIM = 0.3451) indicates a trade-off between aggressive noise suppression and morphology preservation. Morphological analysis showed that Wiener, Median, and Gaussian filtering introduced smaller deviations in extracted burst properties, whereas stronger processing approaches produced larger variations.

Therefore, Wavelet filtering is advantageous for detection-oriented applications that require maximum signal recovery, whereas Wiener filtering provides a more reliable approach for quantitative FRB feature extraction.

4.6 Statistical Validation of Preprocessing Performance

Since each FRB event was processed using all preprocessing approaches, paired non-parametric statistical tests were performed across the complete dataset. The Friedman test demonstrated statistically significant differences among preprocessing methods for all major evaluation metrics.

The obtained Friedman test statistics were 1521.82 for Peak SNR, 1585.81 for Integrated SNR, 2036.78 for Peak Preservation, 2129.28 for SSIM, and 1738.02 for PSNR, with all tests yielding $p < 0.001$.

Subsequent Wilcoxon signed-rank pairwise comparisons with Bonferroni correction confirmed significant differences between multiple preprocessing method combinations. These results demonstrate that the observed performance variations arise from systematic differences between preprocessing strategies rather than random variations within the FRB sample.

	Metric	Statistic	p-value		Peak SNR	Integrated SNR	Peak Preservation (%)	SSIM	PSNR	Total Wins
0	Peak SNR	1521.817623	0.0	Wiener	229	101	196	520	517	1563
1	Integrated SNR	1585.806943	0.0	Median	151	45	323	0	0	519
2	Peak Preservation (%)	2036.776235	0.0	Boxcar	46	195	5	0	0	246
3	SSIM	2129.277170	0.0	Gaussian	76	122	5	0	0	203
4	PSNR	1738.016288	0.0	Wavelet	28	67	6	15	7	123
				Bilateral	5	5	0	0	11	21

Fig(left) - Wilcoxon statistical analysis and the p-value for different metrics , (right) - Wins table across different metrics for all filters through all 535 FRBs.

4.7 Implications for FRB Preprocessing Framework.

The results demonstrate that no single preprocessing method is optimal for all FRB analysis objectives. Wavelet filtering provides maximum signal recovery and is suitable for detection-focused pipelines, while Wiener filtering offers the best compromise between enhancement, image quality, and preservation of intrinsic burst morphology.

Contrast enhancement techniques such as Histogram Equalisation and CLAHE improve visual intensity separation but should not replace denoising algorithms for quantitative FRB analysis, as they primarily redistribute intensity rather than suppress noise.

Therefore, an objective-dependent preprocessing framework is recommended: Wavelet-based enhancement for weak burst detection and Wiener-based processing for accurate morphology and feature extraction. These results highlight the importance of adaptive two-dimensional image-processing approaches for large-scale FRB dynamic spectrum analysis.

4.8 Peak Detection using 2D Wavelet Transform

A 2D Discrete Wavelet Transform (DWT) using the Daubechies-1 (db1) wavelet was applied to the baseline-corrected and frequency-binned waterfall. Soft thresholding of the detail coefficients was performed to suppress background noise while preserving burst-like structures. The denoised waterfall was then reconstructed and the pixel with the maximum intensity was identified as the burst peak.

The processed waterfall is shown in Figure X. The detected peak is marked by a red cross and is located at approximately time index 13 and 600 MHz. The denoising step successfully enhanced the high-intensity burst region, making the burst more distinguishable from the surrounding background noise. The burst appears as a localized bright structure, indicating that the wavelet-based approach effectively preserved the transient signal while reducing noise.

The corresponding time series generated from the denoised waterfall was used to compute the Signal-to-Noise Ratio (SNR), providing a quantitative measure of the improvement achieved after denoising. The detected peak coordinates were subsequently used as the burst location for further analysis.

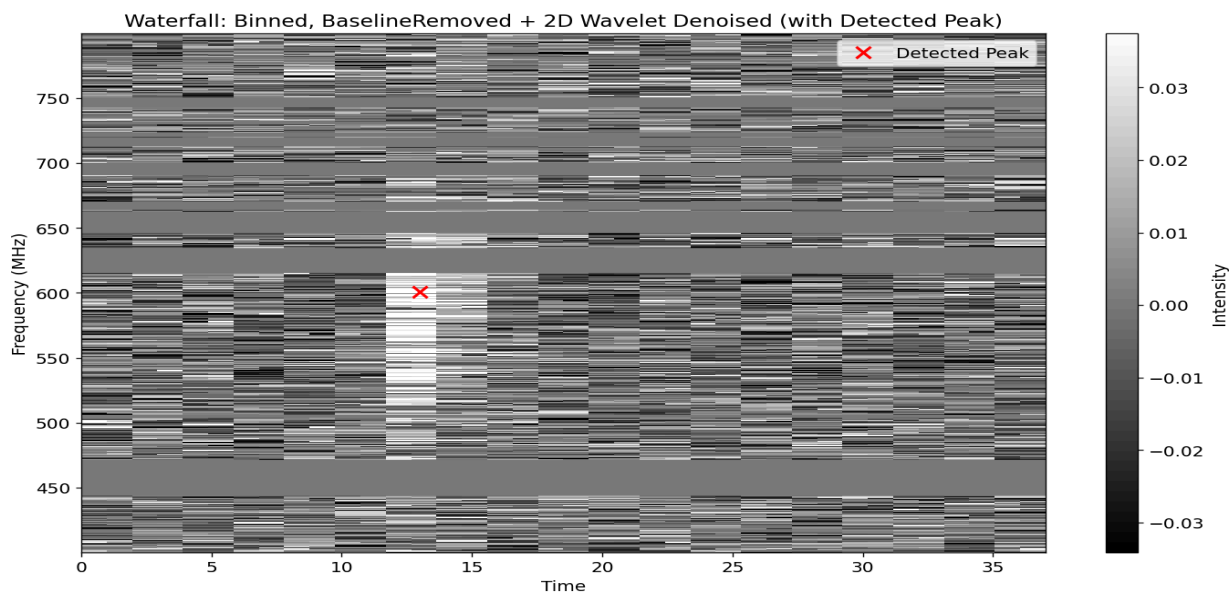


Figure X: Baseline-corrected and frequency-binned waterfall after 2D wavelet denoising. The red cross indicates the detected burst peak.

Following peak detection, the denoised waterfall was analyzed to identify the spatial extent of the burst. A rectangular bounding box was generated around the connected high-intensity region corresponding to the detected burst. As shown in Figure Y, the bounding box successfully encloses the dominant burst structure while separating it from the surrounding background noise.

The detected burst is concentrated around 13–16 ms in time and spans approximately 500–600 MHz in frequency. The highlighted region provides an estimate of the burst duration and frequency bandwidth, enabling subsequent analysis such as burst localization, parameter extraction, and characterization. The use of a bounding box also provides a clear visual representation of the detected burst region, facilitating qualitative verification of the detection results.

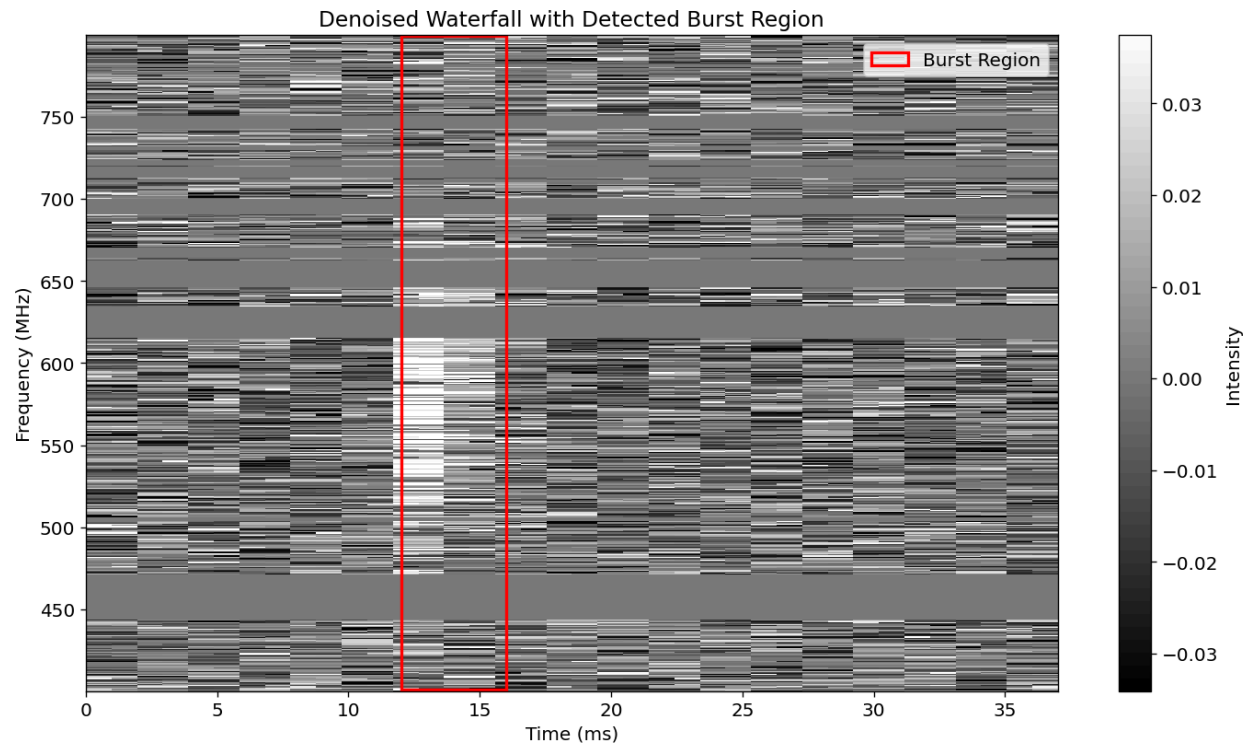


Figure Y: Denoised waterfall with the detected burst region highlighted using a red bounding box.

Section 5: Limitations, Conclusion, & Future Scope

5.1 Limitations

Although the proposed signal processing framework improves the visibility and detection of Fast Radio Bursts (FRBs), several limitations remain.

The performance of the framework is dependent on a number of user-defined parameters, including denoising filter settings, wavelet decomposition levels, threshold values, and frequency binning size. These parameters were selected empirically based on experimental observations and may require retuning for datasets with different noise characteristics or observational conditions. Consequently, the robustness of the pipeline may vary when applied to data acquired from different radio telescopes or under varying interference environments.

While the preprocessing stage effectively suppresses a significant amount of Radio Frequency Interference (RFI), strong broadband or impulsive interference may still persist within the dynamic spectrum. Such residual interference can distort burst morphology, reduce the effective Signal-to-Noise Ratio (SNR), and occasionally lead to inaccurate burst localization or false detections.

The current framework is primarily designed to detect isolated burst structures. FRBs exhibiting multiple peaks, overlapping components, or highly irregular temporal and spectral morphologies may not be fully characterized, which can reduce the accuracy of burst localization and subsequent analysis.

Burst localisation is performed using wavelet-based enhancement followed by thresholding and connected-component analysis. Although this approach successfully identifies candidate burst regions in most cases, the generated bounding boxes may not always precisely correspond to the actual extent of weak, diffuse, or fragmented bursts.

Furthermore, the proposed framework has been evaluated primarily using dynamic spectra from the CHIME/FRB Catalogue. Since observations from different radio telescopes vary in frequency coverage, bandwidth, time resolution, and instrumental characteristics, further validation on multiple datasets is required to assess the generalizability and robustness of the proposed methodology.

5.2 Conclusion

This work presented a reproducible image-processing framework for Fast Radio Burst (FRB) dynamic spectra using data from the CHIME/FRB Open Data Catalogue. Unlike conventional one-dimensional approaches, the proposed framework operates directly on two-dimensional waterfall images, allowing burst morphology to be preserved while reducing instrumental noise and Radio Frequency Interference (RFI). The pipeline integrates baseline correction, adaptive preprocessing, multiple denoising techniques, quantitative evaluation metrics, and automated burst detection within a unified processing workflow.

Comparative analysis across 535 independent FRB events demonstrated that adaptive preprocessing methods consistently outperformed conventional smoothing techniques. Wavelet filtering achieved the highest signal enhancement and noise suppression, making it well suited for burst detection, while Wiener filtering provided the best balance between image quality and preservation of intrinsic burst morphology. Statistical validation using Friedman and Wilcoxon tests confirmed that these improvements were statistically significant and reflected systematic differences between preprocessing strategies rather than random variation.

The implemented 2D wavelet-based burst detection algorithm successfully identified burst peaks and localized candidate burst regions, demonstrating that image-domain processing can effectively enhance burst visibility while providing a practical basis for automated burst characterisation. Although the current bounding-box approach provides an approximate region of interest, it establishes a foundation for further refinement through more advanced segmentation techniques.

The significance of this work extends beyond improving signal quality. By preserving the time-frequency morphology of FRBs, the proposed framework enables more reliable burst characterisation, feature extraction, and quantitative analysis. Such improvements are important for studying burst properties, understanding propagation through the interstellar and intergalactic medium, and supporting the development of robust automated analysis pipelines as the volume of FRB observations continues to grow.

Overall, this study demonstrates that morphology-aware two-dimensional image processing provides a practical and effective approach for enhancing FRB dynamic spectra. The proposed framework offers a scalable foundation for future applications, including automated burst classification, machine learning-based detection, and next-generation FRB search and analysis pipelines.

5.3 Future Scope

Several improvements can further enhance the proposed framework. Adaptive parameter selection techniques can be incorporated to automatically determine optimal threshold values, wavelet decomposition levels, denoising parameters, and frequency binning size, thereby reducing manual intervention and improving robustness across diverse datasets.

Improve burst segmentation by using adaptive thresholding, connected-component refinement, or contour-based detection to generate tighter and more accurate burst boundaries.

The integration of an iterative Dispersion Measure (DM) refinement stage before burst detection could further improve burst alignment, enhance signal coherence, and increase the accuracy of burst localization and characterization.

Future work may also explore the use of machine learning and deep learning techniques for automated burst detection, classification, and false-positive rejection. In addition, extending the framework to accurately detect multiple or overlapping bursts within a single observation would improve its applicability to more complex FRB events.

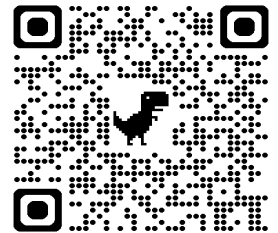
Finally, validating the proposed methodology on observations from multiple radio telescopes and optimizing the implementation for real-time processing would improve its scalability, computational efficiency, and suitability for next-generation large-scale FRB surveys.

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Data & Code Availability:

All source code, processing scripts, and optimisation pipelines developed in this work are publicly available on GitHub at: https://github.com/nischayaj/CHIME_Pipeline_optimization, or you may use the QR code for the same.



Data: <https://www.canfar.net/storage/vault/list/AstroDataCitationDOI/CISTI.CANFAR/21.0007/data/waterfalls/data>